Vittoria De Pellegrini

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Professional Summary

Italian-Brazilian PhD student at King Abdullah University of Science and Technology (KAUST) and Doctoral Researcher with the industry-funded DeepWave Consortium, an initiative that builds machine- and deep-learning workflows for subsurface reservoir characterization. I focus on deep generative AI, especially video diffusion models, to model multi-phase subsurface fluid-flow dynamics. My workflows rely on CUDA-aware, reproducible CPU/GPU parallel computing pipelines running at scale on the KAUST IBEX Supercomputing cluster.

Skills

Programming & Tooling: Python, C++, Visual Studio Code, Git/GitHub, Docker/Docker Hub, Linux terminal workflow
Machine/Deep Learning & Generative AI: Supervised, semi-supervised, and unsupervised learning; linear/logistic regression; random forest; gradient boosting; image segmentation; neural networks; fine-tuning workflows; diffusion models (conditional, latent, video); progressive autoregressive video diffusion models; diffusion transformers; latent space engineering; VAE/VQ-VAE/VQ-GAN (2D/3D)
Python Libraries: PyTorch, TensorFlow, Hugging Face Transformers, Accelerate, Optuna, TensorBoard, Weights & Biases (wandb), NumPy, Pandas, Scikit-Learn, scikit-image, Matplotlib, Seaborn, Folium, Dlisio, Lasio, OPM Flow, pyopmspe11 (SPE11 benchmark), Pickle, JSON, QB Styles
High-Performance Computing: CPU/GPU parallel programming, CUDA job scripting, KAUST IBEX Supercomputing
Productivity & Collaboration: Microsoft Word, Excel, PowerPoint, Overleaf, LaTeX, Slack, Zoom

Professional Experience

DeepWave Research Consortium, KAUST — Doctoral Researcher Jan 2024 – Present
Thuwal, Saudi Arabia
- Build deep generative models for multi-phase subsurface fluid-flow dynamics.
- Build latent conditional diffusion models and latent progressive autoregressive video diffusion models tailored to reservoir simulation data.
- Generate video sequences of key field variables such as CO_2 gas saturation and pressure build-up evolution in the subsurface for CO_2 geological sequestration studies.
- Run large-scale training and inference experiments on the KAUST IBEX multi-GPU cluster through CUDA-aware job scripts and CPU/GPU parallel workflows.

Politecnico di Torino (PoliTO) — Graduate Researcher Feb 2023 – Jul 2023
Turin, Italy
- Developed supervised machine learning models for conventional and advanced well-log prediction.
- Applied the models to wells from the São Francisco and Santos basins in Brazil using PETROBRAS datasets.

Aramco Overseas Company B.V. — Geophysics Intern Sep 2022 – Nov 2022
Delft, Netherlands
- Built a 3D multi-physics synthetic model for a geothermal site in Saudi Arabia.
- Generated surface wave dispersion curves.
- Produced synthetic seismograms.

Education

King Abdullah University of Science and Technology (KAUST) Jan 2024 – Dec 2027 (Expected)
Doctor of Philosophy in Earth Science and Engineering — GPA 3.89 Thuwal, Saudi Arabia
Politecnico di Torino (PoliTO) Oct 2020 – Jul 2023
Master of Science in Petroleum Engineering — Grade 108/110 Turin, Italy
Università degli Studi di Padova (UniPD) Oct 2017 – Jul 2020
Bachelor of Science in Civil Engineering — Grade 110/110 Padua, Italy
Liceo Scientifico Statale Giordano Bruno Sep 2012 – Jun 2017
High School Diploma Venice, Italy
Orewa College Jul 2015 – Sep 2015
Exchange Summer Student Auckland, New Zealand

Languages

Italian — Bilingual • Brazilian Portuguese — Bilingual • English — Advanced • Spanish — Intermediate • Khaliji Arabic — Basic

Journal Publications

V. De Pellegrini, T. Alkhalifah. (2026, October) "LAViG-FLOW: Latent Autoregressive Video Generation for Fluid Flow Simulations." *Journal of Geophysical Research: Machine Learning and Computation*, 3(5), e2026JH001294. DOI: 10.1029/2026JH001294

Conference Publications

V. De Pellegrini, D. Wamriew, T. Alkhalifah. (2025, June) "A Conditional Diffusion Model for CO_2 Monitoring and Forecasting in Heterogeneous Geological Formations." *86th EAGE Annual Conference & Exhibition* (Vol. 2025, pp. 1 - 5). European Association of Geoscientists & Engineers. DOI: 10.3997/2214-4609.202510212

V. De Pellegrini, T. Alkhalifah. (2025, August) "Towards Generative Modeling of CO_2 Geological Storage with Latent Conditional Diffusion Models." *Fifth International Meeting for Applied Geoscience & Energy*. SEG/AAPG, Expanded Abstracts (pp. 1290 - 1294). DOI: 10.1190/image2025-4303381.1

V. De Pellegrini, T. Alkhalifah. (2026, June) "Latent Autoregressive Video Diffusion Models for Fluid Flow Simulations." *87th EAGE Annual Conference & Exhibition* (Vol. 2026, pp. 1 - 5). European Association of Geoscientists & Engineers. DOI: 10.3997/2214-4609.202610173

V. De Pellegrini, T. Alkhalifah. (2026, August) "Physics-Informed Latent Autoregressive Video Diffusion Models for Fluid Flow Simulations." *Sixth International Meeting for Applied Geoscience & Energy*. SEG/AAPG, Expanded Abstracts.

Thesis

V. De Pellegrini. "Development of Supervised Machine Learning Models for the Prediction of Well-Logs & Application on Wells at São Francisco and Santos Basins, Brazil." Master's Thesis, Politecnico di Torino, 2023.